

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		The behaviour of one individual (of a species) {influences / learns} the behaviour of others / Involve interactions between individuals (of the same species) and are used to form structured social groups / owtte	1			1		
		(ii)		{Innate / instinctive} behaviour / initiated by sign stimulus Accept behaviour that is not learned (1) Warns other dogs of {danger / predators} / owtte (1)	1	1		2		
		(iii)		Any two (x1) from: A FAP can be modified by learning / a reflex cannot be learnt (1) FAP can be activated by choice / a reflex action cannot be activated by choice (1) FAP is more complex / ORA (1)	2			2		
	(b)	(i)		(Habituation) is learning to ignore stimuli because they are not followed by reward or punishment/the animal {ceases to respond to/ ignores} a repeated stimulus (1) Saves energy (by not responding to a useless stimulus) (1)	1	1		2		
		(ii)		83.33 / 83.3 / 83 = 2 marks If incorrect award 1 mark for (55-30) / 30 X 100		2		2	2	
		(iii)		increase in the concealment distance (for both colonies) (1) no habituation has occurred (1) concealment distance is greater for the rural colonies (1)			3	3		3
	(c)	(i)		{Examines / shows} brain activity	1			1		
		(ii)		Left cerebral hemisphere / Wernicke's area / left temporal lobe / Broca (1) Neuroplasticity (1) (Undamaged) axons form new connections (1) (new neural pathways / synapses formed) in the right hand side of the brain. (1)		3	1	4		

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		(iii)		{Synaptic pruning / reduction in synapses} of an adult (1) Reduces neuroplasticity (1) Less able to {redevelop language skills / form new pathways} / reduction in the (new) language functioning area (in image 9.5 in comparison to the patient in image 9.4) (1)		2	1	3		
				Question 9 total	6	9	5	20	2	3